

Similarly, the flow map of the system with Hamiltonian $H_2 = U$ is

$$\begin{aligned}\mathbf{Q} &= \mathbf{q}, \\ \mathbf{P} &= \mathbf{p} - h\nabla U(\mathbf{q}).\end{aligned}$$

The composition of these maps simplifies to

$$\begin{aligned}\mathbf{Q} &= \mathbf{q} + h\mathbf{M}^{-1}\mathbf{P}, \\ \mathbf{P} &= \mathbf{p} - h\nabla U(\mathbf{q}),\end{aligned}$$

which is precisely the Symplectic Euler method. Similarly, composing the same two maps in the opposite order gives the adjoint Symplectic Euler method.

Example 2.5 (Verlet is Symplectic) For the Symplectic Euler method $\mathcal{G}_h$ and its adjoint method $\mathcal{G}_h^\dagger$, consider the composition

$$\mathcal{K}_h := \mathcal{G}_{h/2}^\dagger \circ \mathcal{G}_{h/2}.$$

Let

$$\begin{bmatrix} \bar{\mathbf{q}} \\ \bar{\mathbf{p}} \end{bmatrix} = \mathcal{G}_{h/2} \begin{bmatrix} \mathbf{q} \\ \mathbf{p} \end{bmatrix},$$

i.e.,

$$\bar{\mathbf{q}} = \mathbf{q} + (h/2)\mathbf{M}^{-1}\bar{\mathbf{p}}, \quad \bar{\mathbf{p}} = \mathbf{p} - (h/2)\nabla U(\mathbf{q}).$$

and set

$$\begin{bmatrix} \mathbf{Q} \\ \mathbf{P} \end{bmatrix} = \mathcal{G}_{h/2}^\dagger \begin{bmatrix} \bar{\mathbf{q}} \\ \bar{\mathbf{p}} \end{bmatrix},$$

i.e.,

$$\mathbf{Q} = \bar{\mathbf{q}} + (h/2)\mathbf{M}^{-1}\bar{\mathbf{p}}, \quad \mathbf{P} = \bar{\mathbf{p}} - (h/2)\nabla U(\mathbf{Q}).$$

The composition simplifies to

$$\begin{aligned}\bar{\mathbf{p}} &= \mathbf{p} - \frac{h}{2}\nabla U(\mathbf{q}), \\ \mathbf{Q} &= \mathbf{q} + h\mathbf{M}^{-1}\bar{\mathbf{p}}, \\ \mathbf{P} &= \bar{\mathbf{p}} - \frac{h}{2}\nabla U(\mathbf{Q}).\end{aligned}$$